

Report
Of
CivicConnect

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

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Batch: 2025-2029

2026-27

CANDIDATE'S DECLARATION

I, Naman, do hereby declare that the internship report titled “CivicConnect” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All references used in preparing this report have been acknowledged. I affirm that this report represents the work completed during my internship and has not been submitted to any other institute for any degree or diploma requirement. I shall be responsible for any copyright violation arising from material included in this report.

Signature: Naman

Student Name: Naman

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ACKNOWLEDGEMENT

I would like to express my gratitude to YuvaIntern for providing me the opportunity to complete a four week internship in the E-Governance and Digital Services domain. The internship allowed me to work on a practical web development project and apply concepts related to requirement analysis, interface design, backend development, database handling, testing and version control.

I am thankful to everyone who supported and guided me during the internship. The feedback received during the project helped me review the application from a user perspective and improve both the implementation and documentation. I also express my gratitude to my parents for their support and encouragement throughout the internship period.

Signature: Naman

Student Name: Naman

Roll No.: 25SCS1003004379

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4. PROJECT DESCRIPTION

4.1 Introduction

CivicConnect is a community driven e-governance and civic participation web platform developed during a four week remote internship with YuvaIntern. The project was designed to connect local civic issue reporting with community participation in one responsive interface. Residents can join approved communities, report civic issues, suggest local improvements, take part in approved initiatives and view public civic activity through a privacy aware map.

The project progressed from planning and requirement analysis to prototype development, testing, bug fixing and a final review of performance, accessibility, security and privacy. The completed prototype uses Python with Flask for backend logic, SQLite for data storage, HTML, CSS and JavaScript for the interface, React components rendered through JavaScript, and Leaflet with OpenStreetMap for map functionality.

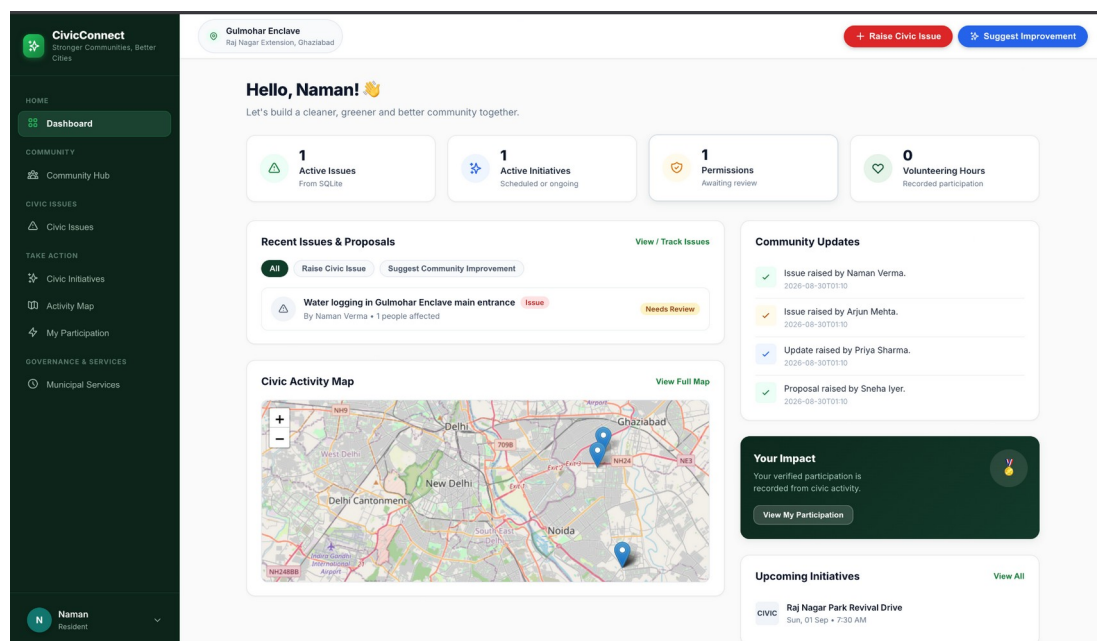


Figure 4.1: CivicConnect dashboard with civic activity and community information.

4.2 Organization Profile

YuvaIntern provided the internship opportunity in the role of Junior Web Developer - E-Governance & Digital Services. The internship was conducted remotely from 05 August 2026 to 02 September 2026 for a duration of four weeks. The assigned work focused on practical web development tasks related to e-governance and digital services, including

interface development, backend integration, database work, testing and project documentation.

4.3 Problem Statement

Citizens often interact with civic services through separate complaint channels, local groups and disconnected information sources. This makes it difficult to connect a reported problem with community discussion, improvement proposals, approved initiatives and public participation. CivicConnect addresses this gap by bringing these activities into one web based prototype while keeping complaint reporting and community improvement suggestions as separate actions.

The project also considers privacy in public civic mapping. Instead of requiring precise GPS access, the prototype uses manual locality search and approximate public activity locations so useful civic information can be displayed without unnecessarily exposing private location details.

The main modules included in the prototype are:

Module	Purpose
Dashboard	Overview of civic activity, communities, initiatives and map information.
Community Hub	Discover approved communities, join multiple communities and switch active community context.
Civic Issues	Report and track local civic problems.
Community Improvements	Submit local improvement ideas separately from complaints.
Civic Initiatives	View approved activities and volunteer for local work.
Civic Activity Map	Search locality manually and view privacy safe public civic activity.
Participation	Record involvement in communities and approved initiatives.

Table 4.1: Main project modules.

4.4 Project Objectives

- Convert requirements for a digital civic service into a practical web application.

- Design a responsive interface that keeps civic issues, community improvements and participation easy to understand.
- Connect frontend actions with Flask routes and SQLite data persistence.
- Test important user flows and fix defects identified during actual use.
- Review performance, accessibility, security and privacy concerns.
- Use Git and GitHub for version control and project submission.

4.5 Scope of the Project

The internship scope covered the design and development of a working web prototype rather than a production government system. The prototype includes resident oriented civic issue reporting, community membership, improvement suggestions, approved civic initiatives, volunteering, participation records and locality based public activity mapping. It also includes authentication and authorization checks, frontend to backend integration and SQLite based persistence.

Large scale production deployment, destructive penetration testing, full accessibility certification, very high concurrency testing and integration with official municipal systems were outside the internship scope. These areas are suitable for future development after additional security, infrastructure and administrative requirements are defined.

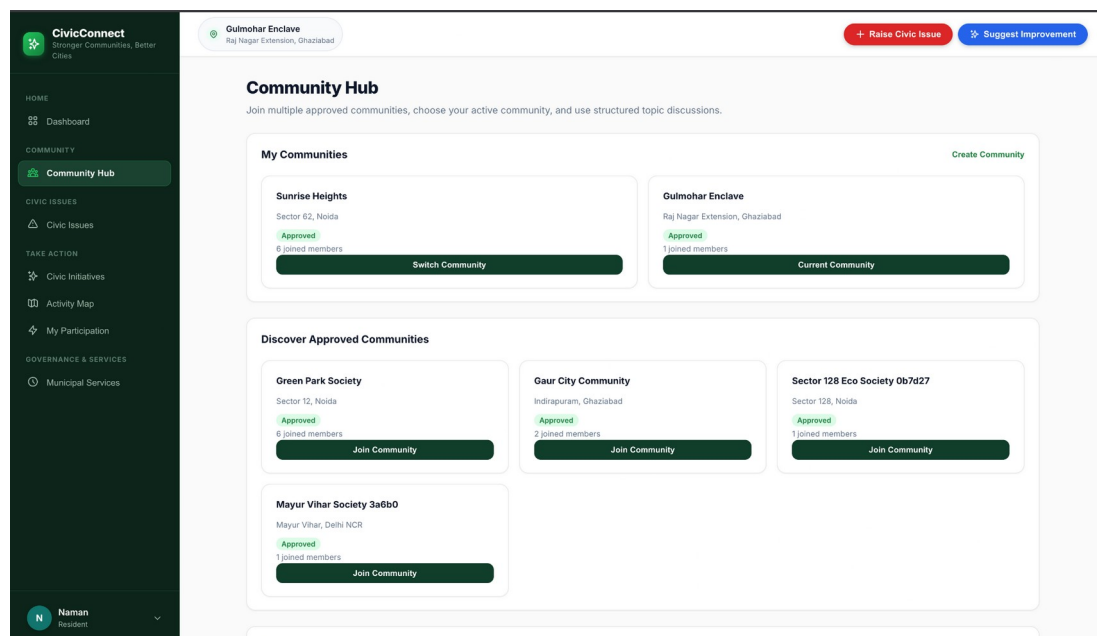


Figure 4.2: Community Hub showing joined and available communities.

4.6 Technologies and Tools Used

Area	Technology / Tool
Frontend	HTML5, CSS3, JavaScript, React through React.createElement
Backend	Python 3.13, Flask 3.1
Database	SQLite3
Map	Leaflet with OpenStreetMap
Version Control	Git and GitHub
Testing	Flask test client, browser testing, Python and JavaScript syntax checks

Table 4.2: Technology stack used in the final prototype.

4.7 System Architecture

CivicConnect follows a simple web application architecture. The user interacts with the browser based frontend. User actions are handled by Flask routes and application logic, which validate requests, apply authorization rules and read or write data in SQLite. The map interface uses Leaflet with OpenStreetMap, while Git and GitHub are used to maintain project versions and prepare the work for submission.

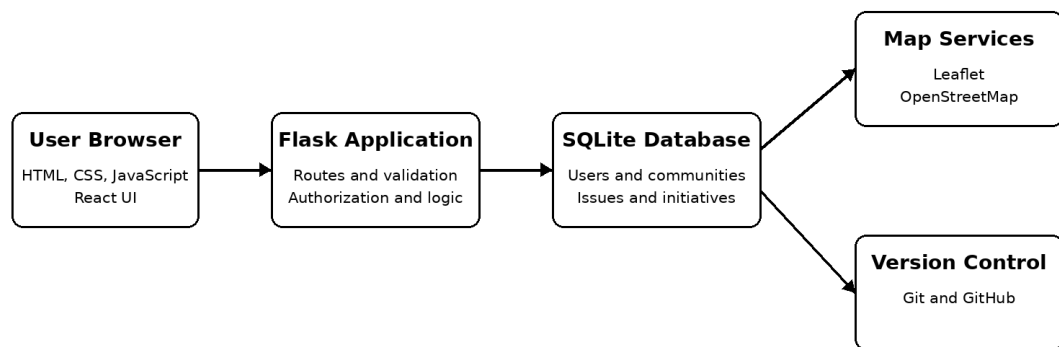


Figure 4.3: High level system architecture of CivicConnect.

A typical user flow is: Dashboard → Community Hub → Civic Issue or Improvement → Review → Approved Initiative → Volunteer → Public Activity Map → Participation

Record. This connects issue reporting with community action while keeping the flow understandable for a resident user.

4.8 Methodology

4.8.1 Week 1: Planning and Requirements

The first week focused on understanding the e-governance problem and defining the project before implementation. Planning covered the main users, required services, functional and non functional requirements, the user journey and a high level technical approach. A four week plan was prepared so development, testing and final review could be handled in separate stages.

During later development, the initial Smart Civic Portal direction evolved into CivicConnect. The final direction placed greater focus on local communities, civic participation and privacy aware activity mapping while keeping the overall e-governance purpose intact.

4.8.2 Week 2: Responsive Prototype Development

Week 2 converted the planning into a working local web prototype. The interface was organized around clear civic actions with persistent navigation, cards, forms, status labels and community sections. The dashboard and Community Hub were built, support for multiple community memberships was added, civic complaints were separated from community improvement proposals, and approved initiatives with volunteering flows were introduced. A searchable activity map using Leaflet and OpenStreetMap was also added.

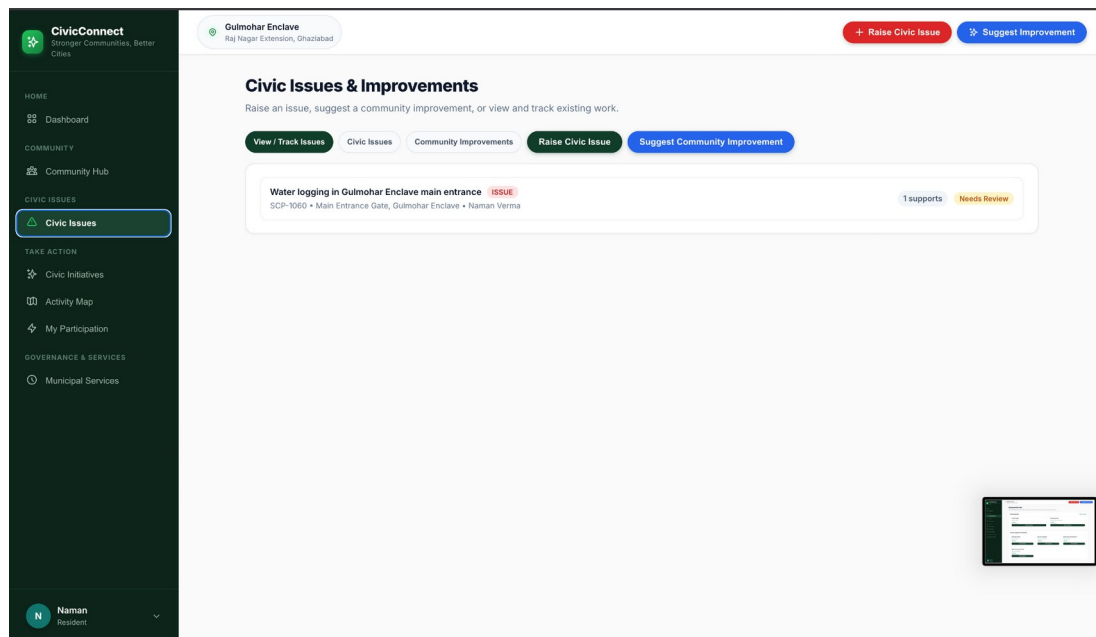


Figure 4.4: Civic issues and community improvement area.

4.8.3 Week 3: Testing and Bug Fixing

Week 3 checked whether the prototype behaved correctly across its main flows. Testing included functional checks, frontend and backend integration, regression testing, authorization checks, database persistence and common negative cases. Manual browser testing was combined with Flask test client checks and syntax validation.

Testing Area	Examples
Functional	Login, issue submission, community join, discussions and volunteering
Integration	Frontend actions, Flask routes and SQLite persistence
Regression	Previously fixed map, modal and JavaScript issues
Authorization	Guest blocking, resident restrictions and protected actions
System Flow	Login through community activity and logout

Table 4.3: Representative testing areas.

The representative Week 3 suite contained 12 test cases and all 12 passed. Four documented defects were fixed and retested. These included a blank screen caused by a JavaScript failure, map and modal overlap, missing map actions and an HTTP 500 error in dashboard aggregation caused by an incorrect database table reference.

4.8.4 Week 4: Final Review and Improvements

The final week reviewed the existing build instead of adding major new features. The audit covered performance, accessibility, security and regression verification. Six SQLite indexes were added for frequently used community, post, initiative, membership and support lookups. Local endpoint measurements showed lower response times after indexing.

Accessibility review covered labels, modal behavior, keyboard use, readable contrast and map limitations. Escape key support was added for closing modals. Security and privacy review covered authorization, SQL injection exposure, session behavior, DOM output and location privacy. Centralized HTML escaping was added for map popup content, and public map data remained approximate rather than exposing precise private locations.

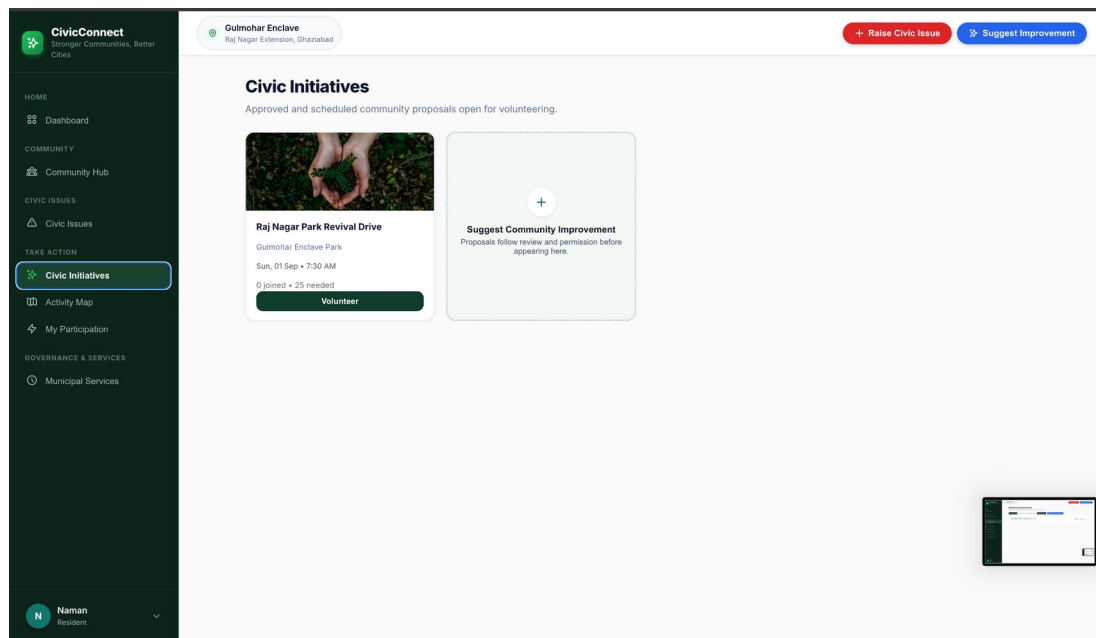


Figure 4.5: Approved civic initiatives and participation options.

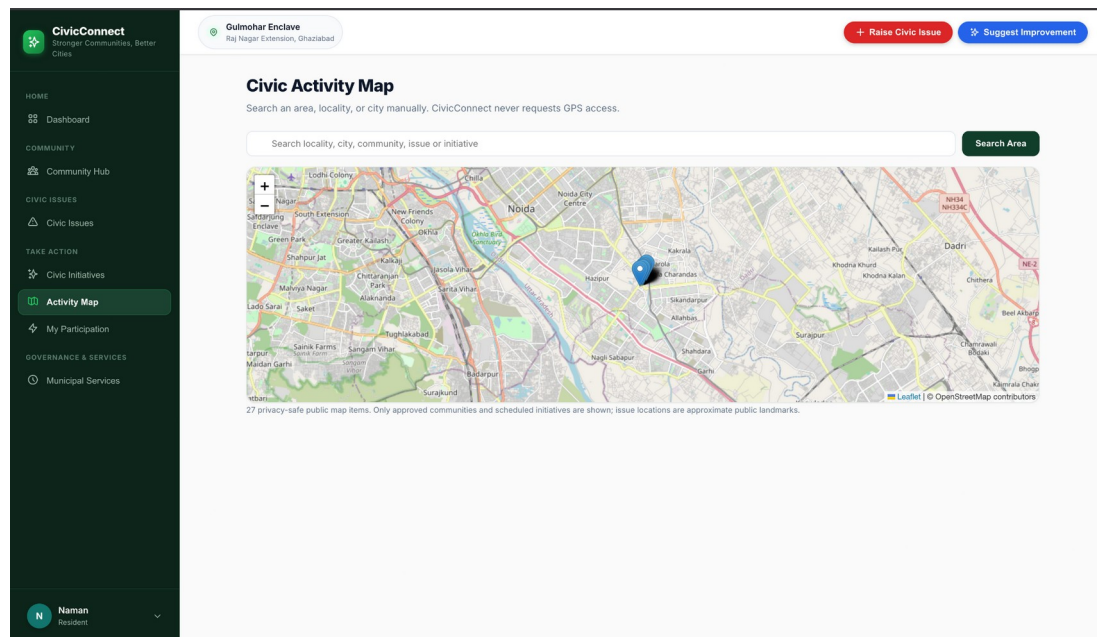


Figure 4.6: Civic activity map with locality based public activity.

4.9 Expected Outcomes

The expected outcome of the internship project was a functional and presentable e-governance web prototype that demonstrates the complete development cycle from planning to final review. The project was expected to provide a responsive civic interface, connected frontend and backend behavior, persistent data storage, clear user flows and practical testing evidence.

- A working responsive prototype for civic issue reporting and community participation.
- Clear separation between complaints and community improvement proposals.
- Support for joining multiple communities and switching active community context.
- Approved civic initiatives and volunteering flows linked to community activity.
- A locality based public activity map designed with location privacy in mind.
- Tested authentication, authorization, database persistence and complete user flows.
- Improved understanding of debugging, testing, performance review, accessibility, security and project organization.

4.10 Certificates of Completion and Communication Proof

The completion certificate and internship offer letter are attached below as proof of the internship engagement, role and four week internship period with YuvaIntern.

4.10.1 Completion Certificate



CERTIFICATE OF EXPERIENCE

To Whomsoever It May Concern

This is to certify that **NAMAN .** successfully completed an internship with **YuvaIntern** in the role of **Junior Web Developer - E-Governance & Digital Services**.

Throughout the tenure of this internship, NAMAN demonstrated consistent dedication, professionalism, and a strong willingness to learn and grow. Their contributions were valued by the entire team at YuvaIntern, and they conducted themselves with integrity and commitment throughout the program.

We commend NAMAN's hard work and wish them continued success in all future endeavors.

Certificate No: YI/2026/184458/409933

Date of Issue: September 02, 2026

FOR AND ON BEHALF OF YUVAINTERN



Kounal Gupta

Founder, [YuvaIntern.com](https://yuvaintern.com)

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4.10.2 Internship Offer Letter / Communication Proof



Dear NAMAN .,

We are pleased to extend to you this formal offer for an Internship position as **Junior Web Developer - E-Governance & Digital Services** at **YuvaIntern**. We were impressed with your background, skills, and enthusiasm, and we believe you will be a valuable addition to our team.

Internship Details:

- **Start Date:** August 05, 2026
- **End Date:** September 02, 2026
- **Duration:** 4 weeks
- **Location:** Remote

Key Responsibilities:

As a **Junior Web Developer - E-Governance & Digital Services** intern, you will work on tasks and projects related to As a Junior Web Developer in the E-Governance & Digital Services sector, you will be responsible for designing, coding, and modifying websites according to client specifications. You will collaborate with cross-functional teams to develop user-friendly interfaces and improve user experience..

Benefits & Learning Opportunities:

Throughout the duration of your internship, you will have the opportunity to:

- Develop practical skills in [relevant skills/tools].
- Gain hands on experience in [specific tasks/projects].

As part of joining our team, you may be required to sign a confidentiality or non-disclosure agreement. You are also expected to comply with all applicable company policies and regulations, which will be provided to you upon commencement.

Kounal Gupta
Founder, YuvaIntern

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1. Python Documentation. Python Software Foundation.
2. Flask Documentation. Pallets Projects.
3. SQLite Documentation. SQLite Project.
4. Leaflet Documentation. Leaflet open source JavaScript library.
5. OpenStreetMap. Open map data and map tiles used for the prototype.
6. Git Documentation and GitHub. Version control and repository hosting.
7. CivicConnect project testing records and weekly internship submissions.